



BITS Pilani
K K Birla Goa Campus

8086/8088 Pins

MICROPROCESSORS & INTERFACING (CS F241)

Dr. Manideepa Mukherjee
Department of CS & IS



I/O Basics

- When data are transferred by **IN** and **OUT** instructions the **I/O address** (port number) is placed on the **address bus**.
- External **IO interface** decode the port number similarly as it decodes the memory address.
- **8 bit** fixed port number appear in address bus **A7-A0** and **16 bit** variable port number (**DX**) appear in address bus **A15-A0**.
- First **256 IO** ports (00H-FFH) are accessed by **fixed** and **variable IO** instructions but **0100H-FFFFH** is only accessed by **variable IO** instructions.
- In PC all **16 address bus** bits are **decoded** with location **0000H** to **03FFH**

Handshaking

I/O devices are typically slower than the microprocessor.

- **Handshaking** is used to **synchronize I/O with the microprocessor**.
- A device indicates that it is **ready for a command** or data (through some I/O pin or port).
- The processor **issues a command** to the device, and the device indicates it is busy (not ready).
- The **I/O device finishes** its task and indicates a ready condition, and the cycle continues.

Handshaking Contd.

There are two basic mechanisms for the processor to service a device.

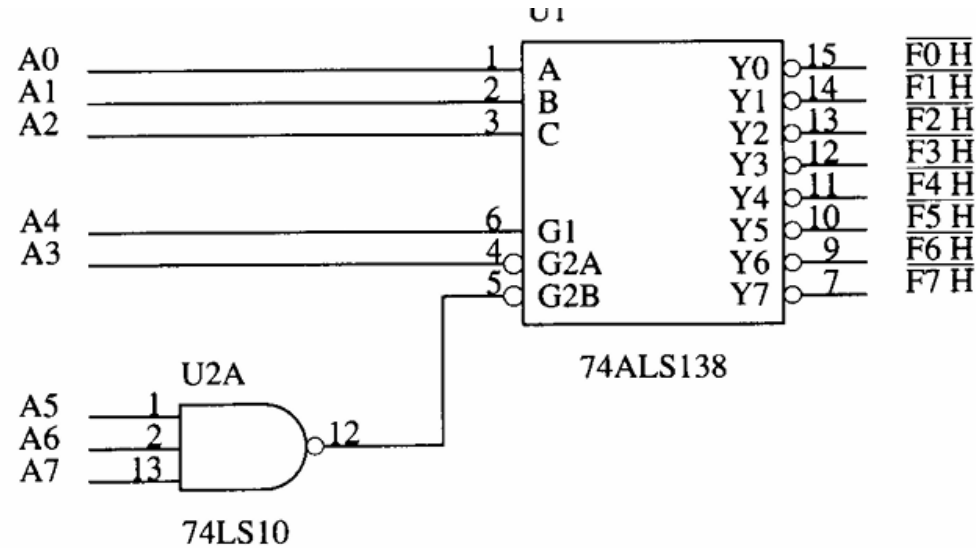
- **Polling: Processor initiated.** Device indicates it is ready by setting some status bit and the processor periodically checks it.
- **Interrupts: Device initiated.** The act of setting a status bit causes an interrupt, and the processor calls an ISR to service the device.

IO Decoding

- It is **similar to Memory address decoding** specially for **memory mapped IO**.
- The decision of using memory mapped IO is often determined by the size of the memory system and the placement of IO devices in the system.
- The main difference between memory decoding and isolated IO decoding is the number of address pins connected to the decoder.
- For memory decoding we use **A31- A0, A23- A0, A19 – A0** but for isolated IO we decode **A15 – A0 (For variable addressing)** and **A7- A0 (For fixed addressing)**.
- **IORC and IOWC** activates the IO devices for read and write operation

IO Decoding: 8 bit IO Port Addresses

- The fixed IO instructions uses 8 bit port address A7-A0 as 0000H – 00FFH.
- The address connections A8-A15 are ignored.
- Embedded systems often uses 8 bit port address. DX register can also address IO ports 00-FFH. If the address is decoded as an 8 bit address it can never include IO that uses 16 bit port address.



IO Decoding: 8 bit IO Port Addresses

```
library ieee;
use ieee.std_logic_1164.all;

entity DECODER_11_11 is

port (
    A7, A6, A5, A4, A3, A2, A1, A0: in STD_LOGIC;
    D0, D1, D2, D3, D4, D5, D6, D7: out STD_LOGIC
);

end;

architecture V1 of DECODER_11_11 is

begin

    D0 <= not( A7 and A6 and A5 and A4 and not A3 and not A2 and not A1 and
               not A0 );
    D1 <= not( A7 and A6 and A5 and A4 and not A3 and not A2 and not A1 and
               A0 );
    D2 <= not( A7 and A6 and A5 and A4 and not A3 and not A2 and A1 and
               not A0 );
```

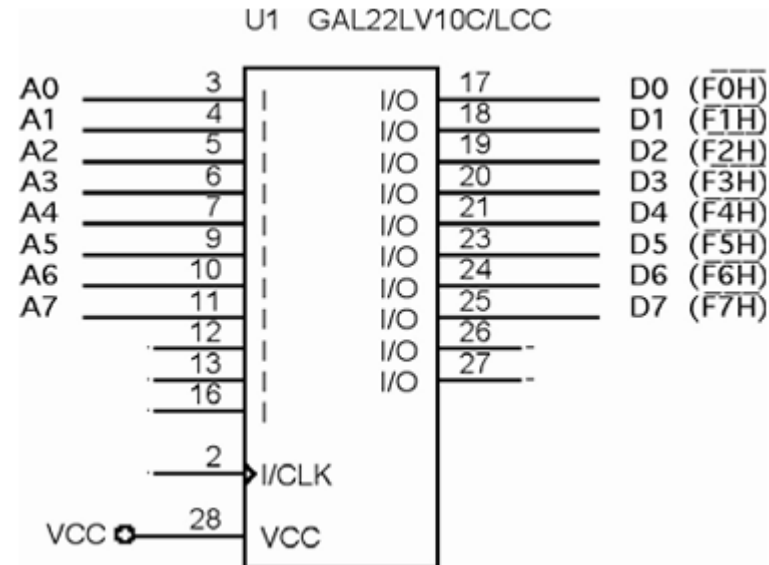
IO Decoding: 8 bit IO Port Addresses

```

D3 <= not( A7 and A6 and A5 and A4 and not A3 and not A2 and A1 and
  A0 );
D4 <= not( A7 and A6 and A5 and A4 and not A3 and A2 and not A1 and
  not A0 );
D5 <= not( A7 and A6 and A5 and A4 and not A3 and A2 and not A1 and
  A0 );
D6 <= not( A7 and A6 and A5 and A4 and not A3 and A2 and A1 and
  not A0 );
D0 <= not( A7 and A6 and A5 and A4 and not A3 and A2 and A1 and
  A0 );

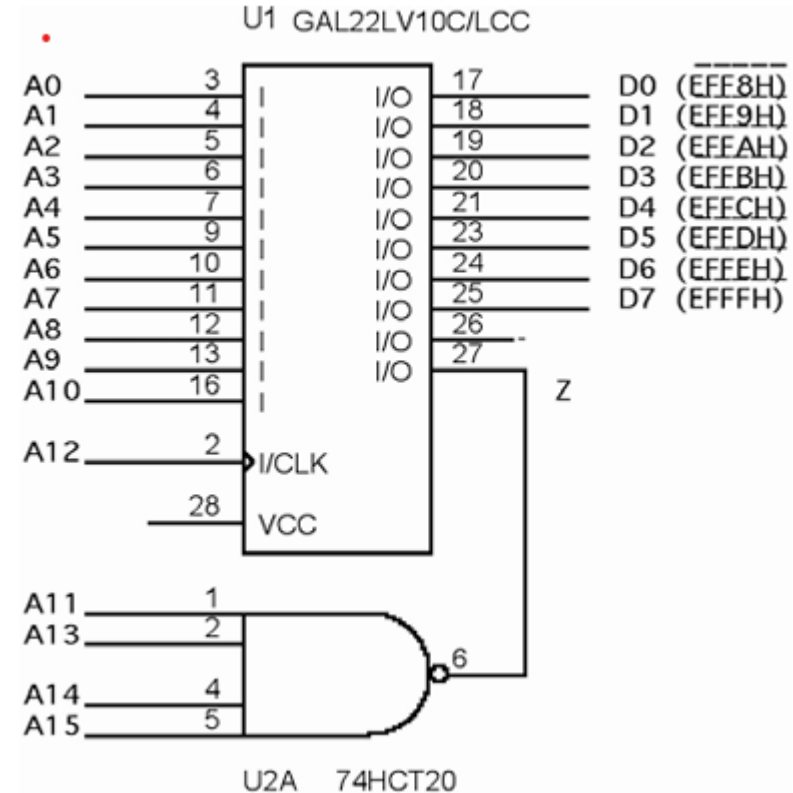
```

end V1;



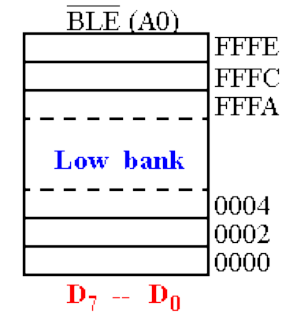
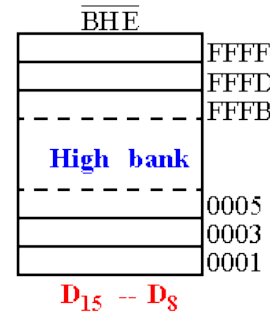
Decoding 16 bit Port Addresses

- Personal computer system use 16 bit I/O addresses.
- In decoding 16 bit IO address the address lines A15-A8 are also decoded.
- A NAND gate and PLD decoder are used to decode IO ports EFF8H-EFFFH.
- NAND gate decoder is used as PLD does not have enough address inputs.
- The OUTPUT of NAND gate is connected to Z input of the PLD and is decoded as part of IO port address.
- The PLD generates Address strobes for IO ports EFF8H- EFFFH.



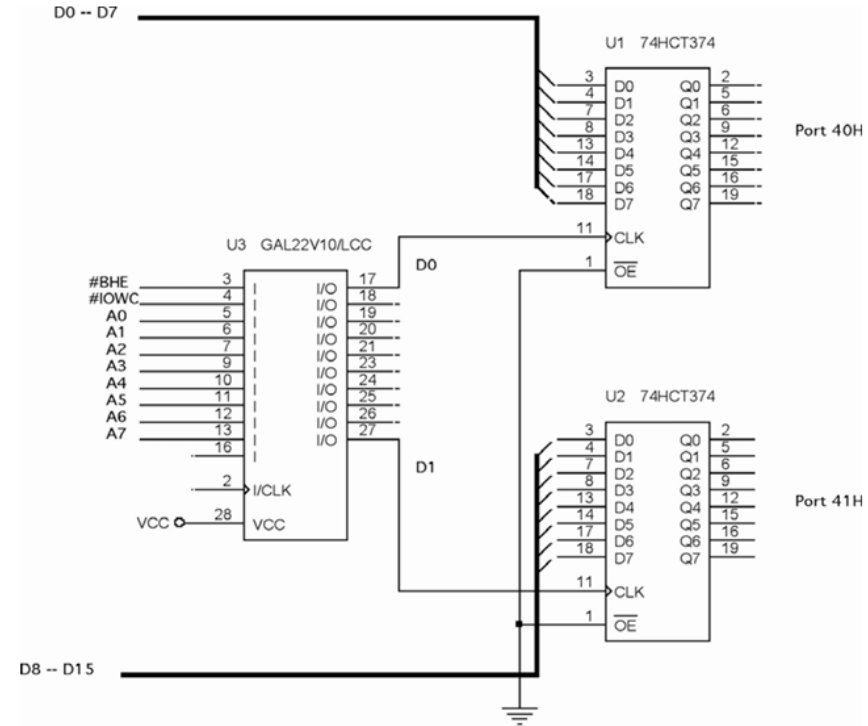
8 Bit and 16 bit Wide IO Ports

- Data transfer to an 8 Bit IO device exist in one of the IO banks in a 16 bit microprocessor such as 80386SX.
- 16 bit port uses two 8 bit ports with 32K different 8 bit ports.
- The IO system in such microprocessor contains two 8 bit memory banks.



8 Bit and 16 bit Wide IO Ports

- 8 bit IO writes require separate write strobe to function correctly.
- IO reads do not require separate read strobes as microprocessor reads only the byte it expects and ignore the other byte.
- The only time that a read can cause problem is when the IO device responds incorrectly to a read operation i.e read from a wrong bank, then there is a need for separate read strobe. It will be discussed later.



```
library ieee;

use ieee.std_logic_1164.all;
entity DECODER_11_14 is

port (
    BHE, IOWC, A7, A6, A5, A4, A3, A2, A1, A0: in STD_LOGIC;
    D0, D1: out STD_LOGIC
);

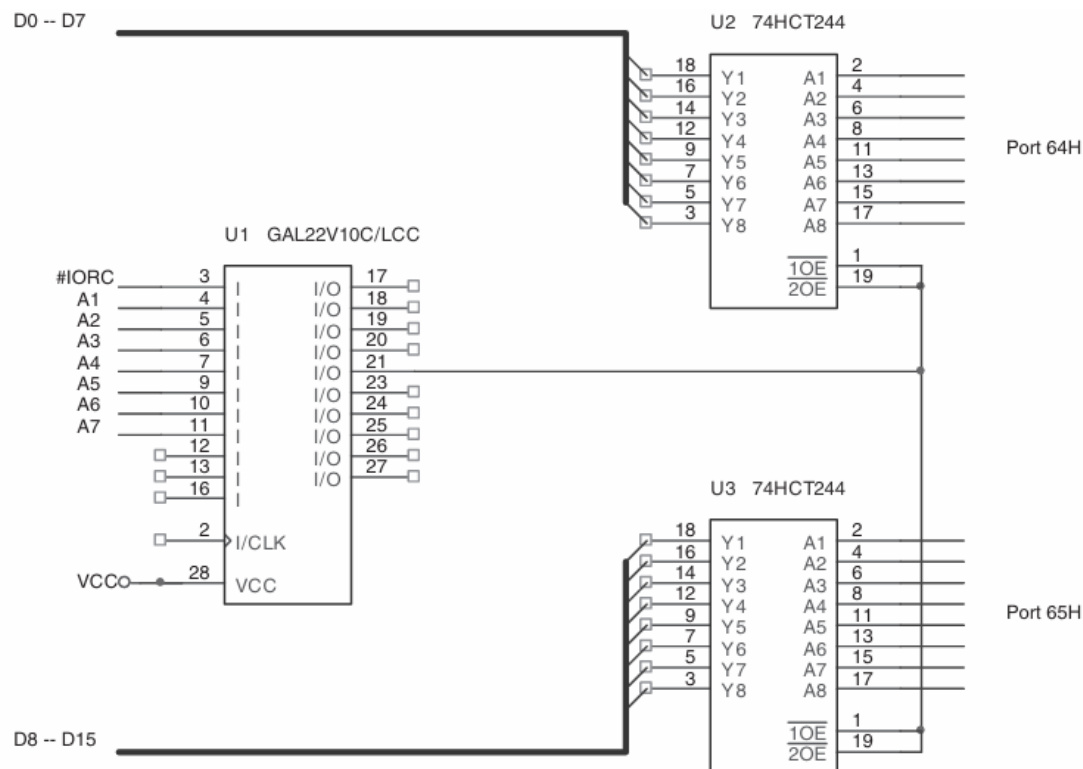
end;

architecture V1 of DECODER_11_14 is

begin

    D0 <= BHE or IOWC or A7 or not A6 or A5 or A4 or A3 or A2 or A1 or A0;
    D1 <= BHE or IOWC or A7 or not A6 or A5 or A4 or A3 or A2 or A1 or
        not A0;

end V1;
```



VHDL Code

```
-- VHDL code for the decoder of Figure 11-15

library ieee;
use ieee.std_logic_1164.all;

entity DECODER_11_15 is

port (
    IORC, A7, A6, A5, A4, A3, A2, A1: in STD_LOGIC;
    D0: out STD_LOGIC
);
```

end;

architecture V1 of DECODER_11_15 is

begin

D0 <= IORC or A7 or not A6 or not A5 or A4 or A3 or not A2 or A1;

end V1;